Contact Model

5.4

Generated by Doxygen 1.9.1

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- file contact_messages.hh

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- file contact_pair.hh

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- file contact_params.hh

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- file contact_surface.hh

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- file line_contact_pair.hh

  Class for a pair of line contact facets for use with contact interaction model.
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  A class to define the interaction type for a pair of contact facets.
- file point_contact_facet.hh

  The contact facet based on the distance to a single point, specifically the vehicle point.
- file point_contact_facet_factory.hh

  Creates a point contact facet from an circular flat plate facet.
- file point_contact_pair.hh

  Class for a pair of point contact facets for use with contact interaction model.
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- file contact.cc

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- file contact_facet.cc

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- file contact_params.cc

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- file contact_surface.cc

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- file contact_surface_factory.cc

  Factory that creates an contact surface, from a surface model.
- file line_contact_facet.cc

  Define LineContactFacet functions.
- file line_contact_facet_factory.cc

  Factory that creates a LineContactFacetFactory from a Cylinder facet and a ContactParams object.
- file line_contact_pair.cc

  LineContactPair class for use with contact interaction model.
- file line_point_contact_pair.cc

  LinePointContactPair class for use with contact interaction model.
- file pair_interaction.cc

  A class to define the interaction type for a pair of contact facets.

- file point_contact_facet.cc

  Define PointContactFacet functions.
- file point_contact_facet_factory.cc

  Factory that creates a PointContactFacet from a FlatPlateCircular facet and a ContactParams object.
- file point_contact_pair.cc

  ContactPair class for use with contact interaction model.
- file spring_pair_interaction.cc

  spring pair interaction for use in the contact model

6.3.1 Detailed Description

Chapter 7

Namespace Documentation

7.1 jeod Namespace Reference

Namespace jeod.

Data Structures

- class Contact

  An base contact class for use in the surface model.
- class ContactFacet

  An contact interaction specific facet for use in the surface model.
- class ContactMessages

  Messages associated with use of the contact model.
- class ContactPair

  An base contact pair class for use in the contact model.
- class ContactParams

  A base class for all contact parameters used in the surface model.
- class ContactSurface

  The contact specific interaction surface, for use with the surface model.
- class ContactSurfaceFactory

  The surface factory that creates an contact specific surface from a general surface.
- class ContactUtils

  Utility string and math functions for the contact model.
- class LineContactFacet

  The contact facet based on the distance to a single point, specifically the vehicle point.
- class LineContactFacetFactory

  Creates a PointContactFacet from an InteractionFacet.
- class LineContactPair

  An point to point contact pair for use in the contact model.
- class LinePointContactPair

  An point to point contact pair for use in the contact model.
- class PairInteraction

  Simple spring contact parameters.
- class PointContactFacet

  The contact facet based on the distance to a single point, specifically the vehicle point.

- class PointContactFacetFactory

  Creates a PointContactFacet from an InteractionFacet.
- class PointContactPair

  An point to point contact pair for use in the contact model.
- class SpringPairInteraction

  Simple spring contact parameters.

7.1.1 Detailed Description

Namespace jeod.

Chapter 8

Data Structure Documentation

8.1 jeod::Contact Class Reference

An base contact class for use in the surface model.

```
#include <contact.hh>
```

Public Member Functions

- Contact ()

    Default Constructor.
- virtual ~Contact ()

    Destructor.
- Contact & operator= (const Contact &)=delete
- Contact (const Contact &)=delete
- void register_contact (ContactFacet ∗facet)

    Register one ContactFacet with all inclusive interactions with other registered ContactFacets.
- void register_contact (ContactFacet ∗∗facets, unsigned int n_facets)

    Register an array of ContactFacets with all inclusive interactions with other registered ContactFacets.
- void register_contact (ContactFacet ∗facet1, ContactFacet ∗facet2)

    Register two facets as a pair.
- void register_contact (ContactFacet ∗∗facets1, unsigned int n_facets1, ContactFacet ∗∗facets2, unsigned int n_facets2)

    Regiser to arrays of facets and create specific pairs between all of them.
- void register_interaction (PairInteraction ∗interaction)

    Register a pair interaction.
- virtual PairInteraction ∗ find_interaction (ContactParams ∗params_1, ContactParams ∗params_2)

    find a PairInteraction baced on a set of ContactParams.
- void initialize_contact (DynManager ∗manager)

    Initialize ContactFacets and the mananger by cleaning up the pair list.
- bool unique_pair (const ContactFacet ∗facet_1, const ContactFacet ∗facet_2)

    Check to see if a pair of facets already exists.
- void check_contact ()

    iterate through contact pairs list then call the appropriate contact resolution functions

Data Fields

- bool active {true}

  toggles contact on and off, true=on false=off
- double contact_limit_factor {}

  factor determines if contact is limited by a muliple of the maximum dimensions of the facets in a pair.

Protected Attributes

- DynManager $*$ dyn_manager {}

  Pointer to the dyn_manager so relstates and be successfully initialized.
- JeodPointerList$<$ ContactPair $>$::type contact_pairs

  list of all possible pairings of contact facets registered with this contact class or derived class
- JeodPointerList$<$ PairInteraction $>$::type pair_interactions

  list of all possible pair interaction types

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2p (jeod, Contact)

8.1.1 Detailed Description

An base contact class for use in the surface model.

Definition at line 91 of file contact.hh.

8.1.2 Constructor & Destructor Documentation

8.1.2.1 Contact() [1/2]

```
jeod::Contact::Contact ( )
```

Default Constructor.

Definition at line 50 of file contact.cc.

References contact_pairs, and pair_interactions.

8.1.2.2 ∼Contact()

```
jeod::Contact::∼Contact ( ) [virtual]
```

Destructor.

Definition at line 62 of file contact.cc.

References contact_pairs, and pair_interactions.

8.1.2.3 Contact() [2/2]

```
jeod::Contact::Contact (
            const Contact & ) [delete]
```

8.1.3 Member Function Documentation

8.1.3.1 check_contact()

```
void jeod::Contact::check_contact ( )
```

iterate through contact pairs list then call the appropriate contact resolution functions

Definition at line 86 of file contact.cc.

References active, and contact_pairs.

8.1.3.2 find_interaction()

```
PairInteraction * jeod::Contact::find_interaction (
            ContactParams * params_1,
            ContactParams * params_2 ) [virtual]
```

find a PairInteraction baced on a set of ContactParams.

Returns

pointer to a PairInteraction

Parameters

in	*params_1*	ContactParams from a ContactFacet
in	*params_2*	ContactParams from a ContactFacet

Definition at line 251 of file contact.cc.

References pair_interactions.

Referenced by jeod::LineContactFacet::create_pair(), and jeod::PointContactFacet::create_pair().

8.1.3.3 initialize_contact()

```
void jeod::Contact::initialize_contact (
            DynManager * manager )
```

Initialize ContactFacets and the mananger by cleaning up the pair list.

Parameters

in,out	*manager*	Dynamics Manager

Definition at line 106 of file contact.cc.

References contact_pairs, jeod::ContactFacet::create_pair(), dyn_manager, jeod::ContactPair::get_subject(), jeod::ContactPair::initialize_relstate(), and unique_pair().

8.1.3.4 JEOD_CLASS_ESTABLISH_FRIENDS2p()

```
jeod::Contact::JEOD_CLASS_ESTABLISH_FRIENDS2p (
            jeod ,
            Contact  )  [private]
```

8.1.3.5 operator=()

```
Contact& jeod::Contact::operator= (
            const Contact &  )  [delete]
```

8.1.3.6 register_contact() [1/4]

```
void jeod::Contact::register_contact (
            ContactFacet ** facets,
            unsigned int nFacets )
```

Register an array of ContactFacets with all inclusive interactions with other registered ContactFacets.

Parameters

`in,out`	*facets*	array of ContactFacets
`in`	*nFacets*	number of ContactFacets in array

Definition at line 184 of file contact.cc.

References register_contact().

8.1.3.7 register_contact() [2/4]

```
void jeod::Contact::register_contact (
            ContactFacet ** facets1,
            unsigned int nFacets1,
            ContactFacet ** facets2,
            unsigned int nFacets2 )
```

Regiser to arrays of facets and create specific pairs between all of them.

Parameters

`in,out`	*facets1*	array of ContactFacets
`in`	*nFacets1*	number of ContactFacets in array
`in,out`	*facets2*	array of ContactFacets
`in`	*nFacets2*	number of ContactFacets in array

Definition at line 220 of file contact.cc.

References register_contact().

8.1.3.8 register_contact() [3/4]

```
void jeod::Contact::register_contact (
            ContactFacet * facet )
```

Register one ContactFacet with all inclusive interactions with other registered ContactFacets.

Parameters

`in,out`	*facet*	ContactFacet

Definition at line 167 of file contact.cc.

References contact_pairs, and jeod::ContactFacet::create_pair().

Referenced by register_contact().

8.1.3.9 register_contact() [4/4]

```
void jeod::Contact::register_contact (
            ContactFacet * facet1,
            ContactFacet * facet2 )
```

Register two facets as a pair.

Parameters

in,out	*facet1*	Contact Facet 1
in,out	*facet2*	Contact Facet 2

Definition at line 197 of file contact.cc.

References contact_pairs, jeod::ContactFacet::create_pair(), and dyn_manager.

8.1.3.10 register_interaction()

```
void jeod::Contact::register_interaction (
            PairInteraction * interaction )
```

Register a pair interaction.

Parameters

in	*interaction*	PairInteraction to add to list

Definition at line 240 of file contact.cc.

References pair_interactions.

8.1.3.11 unique_pair()

```
bool jeod::Contact::unique_pair (
            const ContactFacet * facet_1,
            const ContactFacet * facet_2 )
```

Check to see if a pair of facets already exists.

Returns

bool

Parameters

in,out	*facet_1*	ContactFacet
in,out	*facet_2*	ContactFacet

Definition at line 146 of file contact.cc.

References contact_pairs.

Referenced by initialize_contact().

8.1.4 Field Documentation

8.1.4.1 active

```
bool jeod::Contact::active {true}
```

toggles contact on and off, true=on false=off

trick_units(–)

Definition at line 97 of file contact.hh.

Referenced by check_contact().

8.1.4.2 contact_limit_factor

```
double jeod::Contact::contact_limit_factor {}
```

factor determines if contact is limited by a muliple of the maximum dimensions of the facets in a pair.

trick_units(–)

Definition at line 103 of file contact.hh.

Referenced by jeod::LineContactFacet::create_pair(), and jeod::PointContactFacet::create_pair().

8.1.4.3 contact_pairs

`JeodPointerList<ContactPair>::type jeod::Contact::contact_pairs [protected]`

list of all possible pairings of contact facets registered with this contact class or derived class

trick_io(**)

Definition at line 158 of file contact.hh.

Referenced by check_contact(), Contact(), initialize_contact(), register_contact(), unique_pair(), and ~Contact().

8.1.4.4 dyn_manager

`DynManager* jeod::Contact::dyn_manager {} [protected]`

Pointer to the dyn_manager so relstates and be successfully initialized.

trick_units(–)

Definition at line 152 of file contact.hh.

Referenced by initialize_contact(), and register_contact().

8.1.4.5 pair_interactions

`JeodPointerList<PairInteraction>::type jeod::Contact::pair_interactions [protected]`

list of all possible pair interaction types

trick_io(**)

Definition at line 163 of file contact.hh.

Referenced by Contact(), find_interaction(), register_interaction(), and ~Contact().

The documentation for this class was generated from the following files:

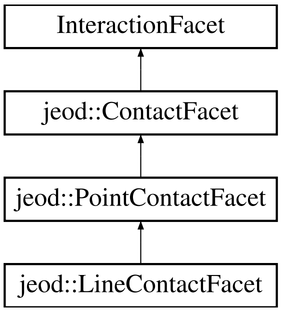
- contact.hh
- contact.cc

8.2 jeod::ContactFacet Class Reference

An contact interaction specific facet for use in the surface model.

`#include <contact_facet.hh>`

Inheritance diagram for jeod::ContactFacet:



Public Member Functions

- ContactFacet ()=default
- ∼ContactFacet () override=default
- ContactFacet & operator= (const ContactFacet &)=delete
- ContactFacet (const ContactFacet &)=delete
- void create_vehicle_point ()

    Create a vehicle point to track the state information of the contact facet.
- virtual void set_max_dimension ()=0

    Calculate the max dimension of the facet for range limit determination.
- std::string get_name () const

    Accessor for name.
- virtual void calculate_torque (double ∗tmp_force)=0

    Calculate the torque acting on the facet in the vehicle structural frame.
- virtual ContactPair ∗ create_pair ()=0

    Overloaded functions that create a ContactPair and pass the address of it to the Contact class for addition to the list of pairs.
- virtual ContactPair ∗ create_pair (ContactFacet ∗target, Contact ∗contact)=0

    Overloaded functions that creates a ContactPair and pass the address of it to the Contact class for addition to the list of pairs.

Data Fields

- bool active {true}

    toggles this contact facet on and off, true=on false=off
- ContactParams ∗ surface_type {}

    Stores the name of surface material that the facet is constructed of.
- DynBody ∗ vehicle_body {}

    DynBody associated with this facet for structural frame information.
- double max_dimension {}

    maximum dimension of the contact facet for use in limiting pair in_contact calls
- double position [3] {}

    position of the facet in vehicle structural frame.
- double normal [3] {}

    normal of the facet relative to the vehicle structural frame.
- const BodyRefFrame ∗ vehicle_point {}

    Vehicle point for relstate calculations.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, ContactFacet)

8.2.1 Detailed Description

An contact interaction specific facet for use in the surface model.

Definition at line 85 of file contact_facet.hh.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 ContactFacet() [1/2]

```
jeod::ContactFacet::ContactFacet ( )  [default]
```

8.2.2.2 ∼ContactFacet()

```
jeod::ContactFacet::∼ContactFacet ( )  [override], [default]
```

8.2.2.3 ContactFacet() [2/2]

```
jeod::ContactFacet::ContactFacet (
            const ContactFacet & )  [delete]
```

8.2.3 Member Function Documentation

8.2.3.1 calculate_torque()

```
virtual void jeod::ContactFacet::calculate_torque (
            double * tmp_force )  [pure virtual]
```

Calculate the torque acting on the facet in the vehicle structural frame.

In: N force from one contact interaction.

Implemented in jeod::PointContactFacet, and jeod::LineContactFacet.

Referenced by jeod::SpringPairInteraction::calculate_forces().

8.2.3.2 create_pair() [1/2]

```
virtual ContactPair* jeod::ContactFacet::create_pair ( )  [pure virtual]
```

Overloaded functions that create a ContactPair and pass the address of it to the Contact class for addition to the list of pairs.

This function is called to create a pair that only contains a subject. Return: – ContactPair that was created

Implemented in jeod::PointContactFacet, and jeod::LineContactFacet.

Referenced by jeod::Contact::initialize_contact(), and jeod::Contact::register_contact().

8.2.3.3 create_pair() [2/2]

```
virtual ContactPair* jeod::ContactFacet::create_pair (
            ContactFacet * target,
            Contact * contact )  [pure virtual]
```

Overloaded functions that creates a ContactPair and pass the address of it to the Contact class for addition to the list of pairs.

This function is called when a subject and target are known. Return: – ContactPair that was created InOut: – target ContactFacet In: – Contact object used to find the pair interaction

Implemented in jeod::PointContactFacet, and jeod::LineContactFacet.

8.2.3.4 create_vehicle_point()

```
void jeod::ContactFacet::create_vehicle_point ( )
```

Create a vehicle point to track the state information of the contact facet.

Create a vehicle point from the base facet position and orientation and store the created vehicle point in the contact facet.

Definition at line 50 of file contact_facet.cc.

References get_name(), normal, position, vehicle_body, and vehicle_point.

Referenced by jeod::LineContactFacetFactory::create_facet(), and jeod::PointContactFacetFactory::create_facet().

8.2.3.5 get_name()

```
std::string jeod::ContactFacet::get_name ( ) const [inline]
```

Accessor for name.

Returns

Point name

Definition at line 173 of file contact_facet.hh.

Referenced by create_vehicle_point().

8.2.3.6 operator=()

```
ContactFacet& jeod::ContactFacet::operator= (
            const ContactFacet & ) [delete]
```

8.2.3.7 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::ContactFacet::p (
            jeod ,
            ContactFacet ) [private]
```

8.2.3.8 set_max_dimension()

```
virtual void jeod::ContactFacet::set_max_dimension ( ) [pure virtual]
```

Calculate the max dimension of the facet for range limit determination.

Implemented in jeod::PointContactFacet, and jeod::LineContactFacet.

8.2.4 Field Documentation

8.2.4.1 active

```
bool jeod::ContactFacet::active {true}
```

toggles this contact facet on and off, true=on false=off

trick_units(–)

Definition at line 91 of file contact_facet.hh.

Referenced by jeod::ContactPair::is_active().

8.2.4.2 max_dimension

```
double jeod::ContactFacet::max_dimension {}
```

maximum dimension of the contact facet for use in limiting pair in_contact calls

trick_units(m)

Definition at line 108 of file contact_facet.hh.

Referenced by jeod::LineContactFacet::create_pair(), jeod::PointContactFacet::create_pair(), jeod::LineContact↩ Facet::set_max_dimension(), and jeod::PointContactFacet::set_max_dimension().

8.2.4.3 normal

```
double jeod::ContactFacet::normal[3] {}
```

normal of the facet relative to the vehicle structural frame.

trick_units(–)

Definition at line 118 of file contact_facet.hh.

Referenced by jeod::LineContactFacetFactory::create_facet(), jeod::PointContactFacetFactory::create_facet(), and create_vehicle_point().

8.2.4.4 position

```
double jeod::ContactFacet::position[3] {}
```

position of the facet in vehicle structural frame.

trick_units(m)

Definition at line 113 of file contact_facet.hh.

Referenced by jeod::LineContactFacetFactory::create_facet(), jeod::PointContactFacetFactory::create_facet(), and create_vehicle_point().

8.2.4.5 surface_type

```
ContactParams* jeod::ContactFacet::surface_type {}
```

Stores the name of surface material that the facet is constructed of.

This information is used to deturmine the contact parameters used when pairs are constructed. trick_units(–)

Definition at line 98 of file contact_facet.hh.

Referenced by jeod::LineContactFacetFactory::create_facet(), jeod::PointContactFacetFactory::create_facet(), jeod::LineContactFacet::create_pair(), and jeod::PointContactFacet::create_pair().

8.2.4.6 vehicle_body

```
DynBody* jeod::ContactFacet::vehicle_body {}
```

DynBody associated with this facet for structural frame information.

trick_units(–)

Definition at line 103 of file contact_facet.hh.

Referenced by jeod::SpringPairInteraction::calculate_forces(), jeod::LineContactFacet::calculate_torque(), jeod::PointContactFacet::calculate_torque(), jeod::LineContactFacetFactory::create_facet(), jeod::PointContactFacetFactory::create_facet(), create_vehicle_point(), and jeod::ContactPair::initialize_relstate().

8.2.4.7 vehicle_point

```
const BodyRefFrame* jeod::ContactFacet::vehicle_point {}
```

Vehicle point for relstate calculations.

trick_units(–)

Definition at line 123 of file contact_facet.hh.

Referenced by jeod::SpringPairInteraction::calculate_forces(), jeod::LineContactFacet::calculate_torque(), jeod::PointContactFacet::calculate_torque(), create_vehicle_point(), jeod::LineContactPair::initialize_pair(), jeod::LinePointContactPair::initialize_pair(), and jeod::PointContactPair::initialize_pair().

The documentation for this class was generated from the following files:

- contact_facet.hh
- contact_facet.cc

8.3 jeod::ContactMessages Class Reference

Messages associated with use of the contact model.

```
#include <contact_messages.hh>
```

Public Member Functions

- ContactMessages ()=delete
- ContactMessages (const ContactMessages &rhs)=delete
- ContactMessages & operator= (const ContactMessages &rhs)=delete

Static Public Attributes

- static const char ∗ initialization_error = "interactions/contact" "initialization_error"

    Associated with errors during initialization of the contact model.
- static const char ∗ runtime_error = "interactions/contact" "runtime_error"

    Associated with errors during the runtime of the contact model.
- static const char ∗ pre_initialization_error = "interactions/contact" "pre_initialization_error"

    Associated with errors during the setup of the system, before runtime.
- static const char ∗ initialization_warns = "interactions/contact" "initialization_warns"

    Associated with warning during initialization of the contact model.
- static const char ∗ runtime_warns = "interactions/contact" "runtime_warns"

    Associated with warnings given at runtime.
- static const char ∗ runtime_inform = "interactions/contact" "runtime_inform"

    Associated with information given at runtime.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 (jeod, ContactMessages)

8.3.1 Detailed Description

Messages associated with use of the contact model.

Definition at line 85 of file contact_messages.hh.

8.3.2 Constructor & Destructor Documentation

8.3.2.1 ContactMessages() [1/2]

```
jeod::ContactMessages::ContactMessages ( )   [delete]
```

8.3.2.2 ContactMessages() [2/2]

```
jeod::ContactMessages::ContactMessages (
            const ContactMessages & rhs )  [delete]
```

8.3.3 Member Function Documentation

8.3.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::ContactMessages::JEOD_CLASS_ESTABLISH_FRIENDS2 (
            jeod ,
            ContactMessages  )  [private]
```

8.3.3.2 operator=()

```
ContactMessages& jeod::ContactMessages::operator= (
            const ContactMessages & rhs )  [delete]
```

8.3.4 Field Documentation

8.3.4.1 initialization_error

```
char const * jeod::ContactMessages::initialization_error = "interactions/contact" "initialization↩
_error"  [static]
```

Associated with errors during initialization of the contact model.

trick_units(--)

Definition at line 95 of file contact_messages.hh.

Referenced by jeod::ContactSurface::allocate_array(), jeod::ContactSurface::allocate_interaction_facet(), jeod↩
::LineContactFacetFactory::create_facet(), jeod::PointContactFacetFactory::create_facet(), and jeod::Contact↩
SurfaceFactory::create_surface().

8.3.4.2 initialization_warns

```
char const * jeod::ContactMessages::initialization_warns = "interactions/contact" "initialization↩
_warns" [static]
```

Associated with warning during initialization of the contact model.

trick_units(–)

Definition at line 110 of file contact_messages.hh.

Referenced by jeod::LineContactFacet::create_pair(), and jeod::PointContactFacet::create_pair().

8.3.4.3 pre_initialization_error

```
char const * jeod::ContactMessages::pre_initialization_error = "interactions/contact" "pre_↩
initialization_error" [static]
```

Associated with errors during the setup of the system, before runtime.

trick_units(–)

Definition at line 103 of file contact_messages.hh.

Referenced by jeod::ContactSurfaceFactory::add_facet_params().

8.3.4.4 runtime_error

```
char const * jeod::ContactMessages::runtime_error = "interactions/contact" "runtime_error"
[static]
```

Associated with errors during the runtime of the contact model.

trick_units(–)

Definition at line 99 of file contact_messages.hh.

8.3.4.5 runtime_inform

```
char const * jeod::ContactMessages::runtime_inform = "interactions/contact" "runtime_inform"
[static]
```

Associated with information given at runtime.

trick_units(–)

Definition at line 121 of file contact_messages.hh.

8.3.4.6 runtime_warns

```
char const * jeod::ContactMessages::runtime_warns = "interactions/contact" "runtime_warns"
[static]
```

Associated with warnings given at runtime.

trick_units(–)

Definition at line 114 of file contact_messages.hh.

The documentation for this class was generated from the following files:

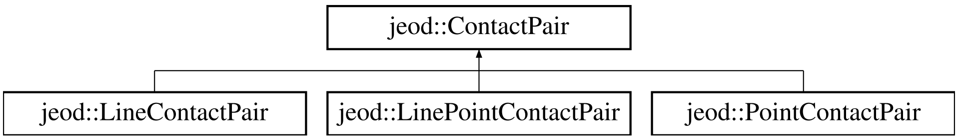
- contact_messages.hh
- contact_messages.cc

8.4 jeod::ContactPair Class Reference

An base contact pair class for use in the contact model.

`#include <contact_pair.hh>`

Inheritance diagram for jeod::ContactPair:



Public Member Functions

- ContactPair ()=default
- virtual ∼ContactPair ()=default
- ContactPair & operator= (const ContactPair &)=delete
- ContactPair (const ContactPair &)=delete
- bool in_range ()

  test whether the pair is in range for interaction
- bool is_active ()

  Determine if contact can occur between the two facets.
- bool is_complete ()

  Determine if the pair has a target facet.
- ContactFacet * get_subject ()

  Determine if the pair has a target facet.
- ContactFacet * get_target ()

  Determine if the pair has a target facet.
- virtual void in_contact ()=0

  Virtual funtion to determine if the pair is in contact.
- virtual void initialize_pair (ContactFacet *subject_facet, ContactFacet *target_facet)=0

  Initialize the contact pair by setting the subject, target, and creating the relstate if possible.
- virtual void initialize_relstate (DynManager *dyn_manager)

  Initialize the relative state between the facets and register with the dynamics manager.
- virtual bool check_tree ()

  Make sure the two contact facets are not on the same mass tree.

Data Fields

- PairInteraction * interaction {}
  Parameters that define the force calculation function between the subjec and target.
- double interaction_distance {}
  rel_state distance at which in_contact should be called

Protected Attributes

- RelativeDerivedState rel_state
  Current relative state between the subject and the target in the subject frame.
- ContactFacet * subject {}
  pointer to the contact facet that is the subject of the associated relative states.
- ContactFacet * target {}
  pointer to the contact facet that is the target of the associated relative states.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, ContactPair)

8.4.1 Detailed Description

An base contact pair class for use in the contact model.

Definition at line 84 of file contact_pair.hh.

8.4.2 Constructor & Destructor Documentation

8.4.2.1 ContactPair() [1/2]

```
jeod::ContactPair::ContactPair ( )  [default]
```

8.4.2.2 ∼ContactPair()

```
virtual jeod::ContactPair::∼ContactPair ( )  [virtual], [default]
```

8.4.2.3 ContactPair() [2/2]

```
jeod::ContactPair::ContactPair (
            const ContactPair & )  [delete]
```

8.4.3 Member Function Documentation

8.4.3.1 check_tree()

```
bool jeod::ContactPair::check_tree ( ) [virtual]
```

Make sure the two contact facets are not on the same mass tree.

Returns

bool

Definition at line 108 of file contact_pair.cc.

References subject, and target.

Referenced by is_active().

8.4.3.2 get_subject()

```
ContactFacet * jeod::ContactPair::get_subject ( )
```

Determine if the pair has a target facet.

Returns

subject ContactFacet

Definition at line 90 of file contact_pair.cc.

References subject.

Referenced by jeod::Contact::initialize_contact().

8.4.3.3 get_target()

```
ContactFacet * jeod::ContactPair::get_target ( )
```

Determine if the pair has a target facet.

Returns

target ContactFacet

Definition at line 99 of file contact_pair.cc.

References target.

8.4.3.4 in_contact()

```
virtual void jeod::ContactPair::in_contact ( )  [pure virtual]
```

Virtual funtion to determine if the pair is in contact.

Contact depends on specific geometry so implementation has to wait for a derived class.

Implemented in jeod::PointContactPair, jeod::LinePointContactPair, and jeod::LineContactPair.

8.4.3.5 in_range()

```
bool jeod::ContactPair::in_range ( )
```

test whether the pair is in range for interaction

Returns

bool

Definition at line 49 of file contact_pair.cc.

References interaction_distance, and rel_state.

8.4.3.6 initialize_pair()

```
virtual void jeod::ContactPair::initialize_pair (
            ContactFacet * subject_facet,
            ContactFacet * target_facet )  [pure virtual]
```

Initialize the contact pair by setting the subject, target, and creating the relstate if possible.

Parameters

in,out	*subject_facet*	subject ContactFacet
in,out	*target_facet*	target ContactFacet

Implemented in jeod::PointContactPair, jeod::LinePointContactPair, and jeod::LineContactPair.

8.4.3.7 initialize_relstate()

```
void jeod::ContactPair::initialize_relstate (
            DynManager * dyn_manager )  [virtual]
```

Initialize the relative state between the facets and register with the dynamics manager.

Initialize the relstate using the DynManager provided by Contact class.

Parameters

in	*dyn_manager*	dynamics manager

Definition at line 129 of file contact_pair.cc.

References rel_state, subject, and jeod::ContactFacet::vehicle_body.

Referenced by jeod::Contact::initialize_contact().

8.4.3.8 is_active()

```
bool jeod::ContactPair::is_active ( )
```

Determine if contact can occur between the two facets.

Returns

bool

Definition at line 64 of file contact_pair.cc.

References jeod::ContactFacet::active, check_tree(), subject, and target.

8.4.3.9 is_complete()

```
bool jeod::ContactPair::is_complete ( )
```

Determine if the pair has a target facet.

Returns

bool

Definition at line 77 of file contact_pair.cc.

References target.

8.4.3.10 operator=()

```
ContactPair& jeod::ContactPair::operator= (
            const ContactPair & ) [delete]
```

8.4.3.11 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::ContactPair::p (
            jeod ,
            ContactPair ) [private]
```

8.4.4 Field Documentation

8.4.4.1 interaction

```
PairInteraction* jeod::ContactPair::interaction {}
```

Parameters that define the force calculation function between the subjec and target.

trick_units(–)

Definition at line 90 of file contact_pair.hh.

Referenced by jeod::LineContactFacet::create_pair(), jeod::PointContactFacet::create_pair(), jeod::LineContactPair::in_contact(), jeod::LinePointContactPair::in_contact(), and jeod::PointContactPair::in_contact().

8.4.4.2 interaction_distance

```
double jeod::ContactPair::interaction_distance {}
```

rel_state distance at which in_contact should be called

trick_units(m)

Definition at line 95 of file contact_pair.hh.

Referenced by jeod::LineContactFacet::create_pair(), jeod::PointContactFacet::create_pair(), and in_range().

8.4.4.3 rel_state

```
RelativeDerivedState jeod::ContactPair::rel_state [protected]
```

Current relative state between the subject and the target in the subject frame.

trick_units(–)

Definition at line 145 of file contact_pair.hh.

Referenced by jeod::LineContactPair::in_contact(), jeod::LinePointContactPair::in_contact(), jeod::PointContactPair::in_contact(), in_range(), jeod::LineContactPair::initialize_pair(), jeod::LinePointContactPair::initialize_pair(), jeod::PointContactPair::initialize_pair(), and initialize_relstate().

8.4.4.4 subject

`ContactFacet* jeod::ContactPair::subject {} [protected]`

pointer to the contact facet that is the subject of the associated relative states.

trick_units(–)

Definition at line 150 of file contact_pair.hh.

Referenced by check_tree(), get_subject(), jeod::LineContactPair::initialize_pair(), jeod::LinePointContactPair↩ ::initialize_pair(), jeod::PointContactPair::initialize_pair(), initialize_relstate(), and is_active().

8.4.4.5 target

`ContactFacet* jeod::ContactPair::target {} [protected]`

pointer to the contact facet that is the target of the associated relative states.

trick_units(–)

Definition at line 155 of file contact_pair.hh.

Referenced by check_tree(), get_target(), jeod::LineContactPair::initialize_pair(), jeod::LinePointContactPair↩ ::initialize_pair(), jeod::PointContactPair::initialize_pair(), is_active(), and is_complete().

The documentation for this class was generated from the following files:

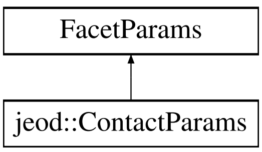
- contact_pair.hh
- contact_pair.cc

8.5 jeod::ContactParams Class Reference

A base class for all contact parameters used in the surface model.

`#include <contact_params.hh>`

Inheritance diagram for jeod::ContactParams:



Public Member Functions

- ContactParams ()
  Default Constructor.
- ∼ContactParams () override=default
- ContactParams & operator= (const ContactParams &)=delete
- ContactParams (const ContactParams &)=delete

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, ContactParams)

8.5.1 Detailed Description

A base class for all contact parameters used in the surface model.

Definition at line 81 of file contact_params.hh.

8.5.2 Constructor & Destructor Documentation

8.5.2.1 ContactParams() [1/2]

```
jeod::ContactParams::ContactParams ( )
```

Default Constructor.

Definition at line 42 of file contact_params.cc.

8.5.2.2 ∼ContactParams()

```
jeod::ContactParams::~ContactParams ( )  [override], [default]
```

8.5.2.3 ContactParams() [2/2]

```
jeod::ContactParams::ContactParams (
            const ContactParams & )  [delete]
```

8.5.3 Member Function Documentation

8.5.3.1 operator=()

```
ContactParams& jeod::ContactParams::operator= (
            const ContactParams & ) [delete]
```

8.5.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::ContactParams::p (
            jeod ,
            ContactParams ) [private]
```

The documentation for this class was generated from the following files:

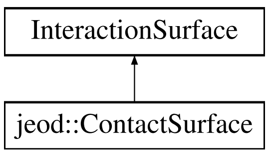
- contact_params.hh
- contact_params.cc

8.6 jeod::ContactSurface Class Reference

The contact specific interaction surface, for use with the surface model.

```
#include <contact_surface.hh>
```

Inheritance diagram for jeod::ContactSurface:



Public Member Functions

- ContactSurface ()

    Default Constructor.
- ∼ContactSurface () override

    Destructor.
- ContactSurface & operator= (const ContactSurface &)=delete
- ContactSurface (const ContactSurface &)=delete
- void allocate_array (unsigned int size) override

    Allocates an array of ContactFacet pointers, of the size indicated by the input variable.
- void allocate_interaction_facet (Facet ∗facet, InteractionFacetFactory ∗factory, FacetParams ∗params, unsigned int index) override

    Creates an interaction facet or more accurately a contact facet from a basic facet and set of parameters.
- virtual void collect_forces_torques ()

    collect the forces and torques from all the facets in this contact surface

Data Fields

- ContactFacet ** contact_facets {}

    An array of pointers to contact interaction facets.
- unsigned int facets_size {}

    Size of the contact_facets array.
- double contact_force [3] {}

    Total Force due to contact, resulting from all plates combined.
- double contact_torque [3] {}

    Total Torque due to contact, resulting from all plates combined.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, ContactSurface)

8.6.1 Detailed Description

The contact specific interaction surface, for use with the surface model.

Definition at line 84 of file contact_surface.hh.

8.6.2 Constructor & Destructor Documentation

8.6.2.1 ContactSurface() [1/2]

```
jeod::ContactSurface::ContactSurface ( )
```

Default Constructor.

Definition at line 56 of file contact_surface.cc.

8.6.2.2 ~ContactSurface()

```
jeod::ContactSurface::~ContactSurface ( )   [override]
```

Destructor.

Definition at line 64 of file contact_surface.cc.

References contact_facets, and facets_size.

8.6.2.3 ContactSurface() [2/2]

```
jeod::ContactSurface::ContactSurface (
            const ContactSurface & )  [delete]
```

8.6.3 Member Function Documentation

8.6.3.1 allocate_array()

```
void jeod::ContactSurface::allocate_array (
            unsigned int size )  [override]
```

Allocates an array of ContactFacet pointers, of the size indicated by the input variable.

Parameters

in	*size*	The size of the needed array Units: cnt:

Definition at line 75 of file contact_surface.cc.

References contact_facets, facets_size, and jeod::ContactMessages::initialization_error.

8.6.3.2 allocate_interaction_facet()

```
void jeod::ContactSurface::allocate_interaction_facet (
            Facet * facet,
            InteractionFacetFactory * factory,
            FacetParams * params,
            unsigned int index )  [override]
```

Creates an interaction facet or more accurately a contact facet from a basic facet and set of parameters.

Parameters

in	*facet*	The basic facet used to create the interaction facet
in	*factory*	The factory used to create the interaction facet
in	*params*	The contact params used to create the interaction facet
in	*index*	Where the new interaction facet will be placed in the contact_facets array Units: cnt

Definition at line 108 of file contact_surface.cc.

References contact_facets, facets_size, and jeod::ContactMessages::initialization_error.

8.6.3.3 collect_forces_torques()

```
void jeod::ContactSurface::collect_forces_torques ( )  [virtual]
```

collect the forces and torques from all the facets in this contact surface

Definition at line 177 of file contact_surface.cc.

References contact_facets, contact_force, contact_torque, and facets_size.

8.6.3.4 operator=()

```
ContactSurface& jeod::ContactSurface::operator= (
            const ContactSurface &  )  [delete]
```

8.6.3.5 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::ContactSurface::p (
            jeod ,
            ContactSurface  )  [private]
```

8.6.4 Field Documentation

8.6.4.1 contact_facets

```
ContactFacet** jeod::ContactSurface::contact_facets {}
```

An array of pointers to contact interaction facets.

trick_units(–)

Definition at line 95 of file contact_surface.hh.

Referenced by allocate_array(), allocate_interaction_facet(), collect_forces_torques(), and ∼ContactSurface().

8.6.4.2 contact_force

```
double jeod::ContactSurface::contact_force[3] {}
```

Total Force due to contact, resulting from all plates combined.

trick_units(N)

Definition at line 105 of file contact_surface.hh.

Referenced by collect_forces_torques().

8.6.4.3 contact_torque

```
double jeod::ContactSurface::contact_torque[3] {}
```

Total Torque due to contact, resulting from all plates combined.

trick_units(N/m)

Definition at line 110 of file contact_surface.hh.

Referenced by collect_forces_torques().

8.6.4.4 facets_size

```
unsigned int jeod::ContactSurface::facets_size {}
```

Size of the contact_facets array.

trick_units(count)

Definition at line 100 of file contact_surface.hh.

Referenced by allocate_array(), allocate_interaction_facet(), collect_forces_torques(), and ∼ContactSurface().

The documentation for this class was generated from the following files:

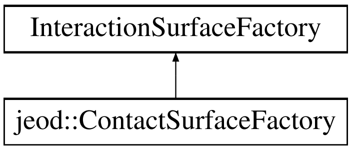
- contact_surface.hh
- contact_surface.cc

8.7 jeod::ContactSurfaceFactory Class Reference

The surface factory that creates an contact specific surface from a general surface.

```
#include <contact_surface_factory.hh>
```

Inheritance diagram for jeod::ContactSurfaceFactory:



Public Member Functions

- ContactSurfaceFactory ()

  Default Constructor.
- ∼ContactSurfaceFactory () override=default
- ContactSurfaceFactory & operator= (const ContactSurfaceFactory &)=delete
- ContactSurfaceFactory (const ContactSurfaceFactory &)=delete
- void create_surface (SurfaceModel ∗surface, InteractionSurface ∗inter_surface) override

  Creates an interaction surface, in the inter_surface parameter, from the given SurfaceModel.
- void add_facet_params (FacetParams ∗to_add) override

  Add a named set of facet params to the surface factory.

Protected Attributes

- PointContactFacetFactory point_contact_facet_factory

  A factory that can create a point contact facet from a circular flat plate.
- LineContactFacetFactory line_contact_facet_factory

  A factory that can create a line contact facet from a cylinder.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, ContactSurfaceFactory)

8.7.1 Detailed Description

The surface factory that creates an contact specific surface from a general surface.

Used with the surface model.

Definition at line 86 of file contact_surface_factory.hh.

8.7.2 Constructor & Destructor Documentation

8.7.2.1 ContactSurfaceFactory() [1/2]

```
jeod::ContactSurfaceFactory::ContactSurfaceFactory ( )
```

Default Constructor.

Definition at line 48 of file contact_surface_factory.cc.

References line_contact_facet_factory, and point_contact_facet_factory.

8.7.2.2 ∼ContactSurfaceFactory()

```
jeod::ContactSurfaceFactory::~ContactSurfaceFactory ( ) [override], [default]
```

8.7.2.3 ContactSurfaceFactory() [2/2]

```
jeod::ContactSurfaceFactory::ContactSurfaceFactory (
            const ContactSurfaceFactory & ) [delete]
```

8.7.3 Member Function Documentation

8.7.3.1 add_facet_params()

```
void jeod::ContactSurfaceFactory::add_facet_params (
            FacetParams * to_add ) [override]
```

Add a named set of facet params to the surface factory.

Intended to be used when an contact specific surface is created, to convert a basic facet to an contact interaction facet. This MUST be a parameter inheriting from ContactParam, or the function will fail and send a failure message

Parameters

in	*to_add*	The facet parameters to add

Definition at line 176 of file contact_surface_factory.cc.

References jeod::ContactMessages::pre_initialization_error.

8.7.3.2 create_surface()

```
void jeod::ContactSurfaceFactory::create_surface (
            SurfaceModel * surface,
            InteractionSurface * inter_surface ) [override]
```

Creates an interaction surface, in the inter_surface parameter, from the given SurfaceModel.

The InteractionSurfaceFactory should contain all necessary InteractionFacetFactories and FacetParams already

Parameters

in	*surface*	The surface model used to create the interaction surface
out	*inter_surface*	Where the interaction surface will be produced

Definition at line 65 of file contact_surface_factory.cc.

References jeod::ContactMessages::initialization_error.

8.7.3.3 operator=()

```
ContactSurfaceFactory& jeod::ContactSurfaceFactory::operator= (
            const ContactSurfaceFactory & ) [delete]
```

8.7.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::ContactSurfaceFactory::p (
            jeod ,
            ContactSurfaceFactory ) [private]
```

8.7.4 Field Documentation

8.7.4.1 line_contact_facet_factory

```
LineContactFacetFactory jeod::ContactSurfaceFactory::line_contact_facet_factory [protected]
```

A factory that can create a line contact facet from a cylinder.

trick_units(--)

Definition at line 115 of file contact_surface_factory.hh.

Referenced by ContactSurfaceFactory().

8.7.4.2 point_contact_facet_factory

`PointContactFacetFactory jeod::ContactSurfaceFactory::point_contact_facet_factory [protected]`

A factory that can create a point contact facet from a circular flat plate.

trick_units(–)

Definition at line 110 of file contact_surface_factory.hh.

Referenced by ContactSurfaceFactory().

The documentation for this class was generated from the following files:

- contact_surface_factory.hh
- contact_surface_factory.cc

8.8 jeod::ContactUtils Class Reference

Utility string and math functions for the contact model.

`#include <contact_utils.hh>`

Static Public Member Functions

- static int create_relstate_name (char ∗name1, char ∗name2, char ∗∗out_str)
  create a name for a relstate out of two facet names
- static int dist_line_segments (double p1[3], double p2[3], double p3[3], double p4[3], double ∗pa, double ∗pb)
  calculate the closest points between two line segments

8.8.1 Detailed Description

Utility string and math functions for the contact model.

Definition at line 72 of file contact_utils.hh.

8.8.2 Member Function Documentation

8.8.2.1 create_relstate_name()

```
int jeod::ContactUtils::create_relstate_name (
            char * name1,
            char * name2,
            char ** out_str ) [inline], [static]
```

create a name for a relstate out of two facet names

Returns

char∗∗

Parameters

in	*name1*	name of first contact facet
in	*name2*	name of second contact facet
in,out	*out_str*	output name for the relstate

Definition at line 93 of file contact_utils_inline.hh.

8.8.2.2 dist_line_segments()

```
int jeod::ContactUtils::dist_line_segments (
            double p1[3],
            double p2[3],
            double p3[3],
            double p4[3],
            double * pa,
            double * pb )  [inline], [static]
```

calculate the closest points between two line segments

Returns

success

Parameters

in	*p1*	vector to one point of the first line seg Units: M
in	*p2*	vector to one point of the first line seg Units: M
in	*p3*	vector to one point of the second line seg Units: M
in	*p4*	vector to one point of the second line seg Units: M
out	*pa*	vector to close_point on line 1 Units: M
out	*pb*	vector to close_point on line 2 Units: M

Definition at line 118 of file contact_utils_inline.hh.

Referenced by jeod::LineContactPair::in_contact(), and jeod::LinePointContactPair::in_contact().

The documentation for this class was generated from the following files:

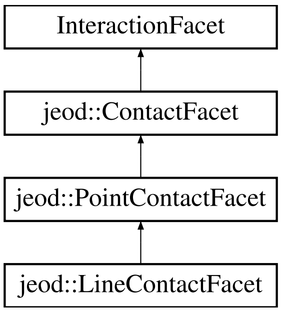
- contact_utils.hh
- contact_utils_inline.hh

8.9 jeod::LineContactFacet Class Reference

The contact facet based on the distance to a single point, specifically the vehicle point.

```
#include <line_contact_facet.hh>
```

Inheritance diagram for jeod::LineContactFacet:



Public Member Functions

- LineContactFacet ()

  Default constructor.
- ∼LineContactFacet () override=default
- LineContactFacet & operator= (const LineContactFacet &)=delete
- LineContactFacet (const LineContactFacet &)=delete
- ContactPair ∗ create_pair () override

  Overloaded functions that create a ContactPair and pass the address of it to the Contact class for addition to the list of pairs.
- ContactPair ∗ create_pair (ContactFacet ∗target, Contact ∗contact) override

  Overloaded functions that creates a ContactPair and pass the address of it to the Contact class for addition to the list of pairs.
- void set_max_dimension () override

  calculate the max dimension of the facet for range limit determination.
- void calculate_torque (double ∗tmp_force) override

  Calculate the torque generated on the vehicle by the facet.
- void calculate_contact_point (double nvec[3]) override

  Find the point on the surface that coorisponds to the closest point on line segments using the radius value.

Data Fields

- double length {}

  length of the line along the vehicle point x axis.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, LineContactFacet)

8.9.1 Detailed Description

The contact facet based on the distance to a single point, specifically the vehicle point.

In effect this represents a sphere.

Definition at line 87 of file line_contact_facet.hh.

8.9.2 Constructor & Destructor Documentation

8.9.2.1 LineContactFacet() [1/2]

```
jeod::LineContactFacet::LineContactFacet ( )
```

Default constructor.

Definition at line 52 of file line_contact_facet.cc.

8.9.2.2 ∼LineContactFacet()

```
jeod::LineContactFacet::~LineContactFacet ( ) [override], [default]
```

8.9.2.3 LineContactFacet() [2/2]

```
jeod::LineContactFacet::LineContactFacet (
            const LineContactFacet & ) [delete]
```

8.9.3 Member Function Documentation

8.9.3.1 calculate_contact_point()

```
void jeod::LineContactFacet::calculate_contact_point (
            double nvec[3] ) [override], [virtual]
```

Find the point on the surface that coorisponds to the closest point on line segments using the radius value.

Parameters

in	*nvec*	vector between line points

Reimplemented from jeod::PointContactFacet.

Definition at line 159 of file line_contact_facet.cc.

References jeod::PointContactFacet::contact_point, length, and jeod::PointContactFacet::radius.

Referenced by jeod::LineContactPair::in_contact(), and jeod::LinePointContactPair::in_contact().

8.9.3.2 calculate_torque()

```
void jeod::LineContactFacet::calculate_torque (
            double * tmp_force )  [override], [virtual]
```

Calculate the torque generated on the vehicle by the facet.

Assumes that the force is in the vehicle structural frame, but that close_point is not.

Parameters

in	*tmp_force*	force from one contact interaction. Units: N

Implements jeod::ContactFacet.

Definition at line 212 of file line_contact_facet.cc.

References jeod::PointContactFacet::contact_point, jeod::ContactFacet::vehicle_body, and jeod::ContactFacet↩
::vehicle_point.

8.9.3.3 create_pair() [1/2]

```
ContactPair * jeod::LineContactFacet::create_pair ( )  [override], [virtual]
```

Overloaded functions that create a ContactPair and pass the address of it to the Contact class for addition to the list of pairs.

This function is called to create a pair that only contains a subject.

Returns

ContactPair that was created

Implements jeod::ContactFacet.

Definition at line 64 of file line_contact_facet.cc.

References jeod::LineContactPair::initialize_pair().

8.9.3.4 create_pair() [2/2]

```
ContactPair * jeod::LineContactFacet::create_pair (
            ContactFacet * target,
            Contact * contact ) [override], [virtual]
```

Overloaded functions that creates a ContactPair and pass the address of it to the Contact class for addition to the list of pairs.

This function is called when a subject and target are known.

Returns

ContactPair that was created

Parameters

in,out	*target*	target ContactFacet
in	*contact*	Contact object used to find the pair interaction

Implements jeod::ContactFacet.

Definition at line 83 of file line_contact_facet.cc.

References jeod::Contact::contact_limit_factor, jeod::Contact::find_interaction(), jeod::ContactMessages::initialization_warns, jeod::LineContactPair::initialize_pair(), jeod::LinePointContactPair::initialize_pair(), jeod::ContactPair::interaction, jeod::ContactPair::interaction_distance, jeod::ContactFacet::max_dimension, and jeod::ContactFacet::surface_type.

8.9.3.5 operator=()

```
LineContactFacet& jeod::LineContactFacet::operator= (
            const LineContactFacet & ) [delete]
```

8.9.3.6 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::LineContactFacet::p (
            jeod ,
            LineContactFacet ) [private]
```

8.9.3.7 set_max_dimension()

```
void jeod::LineContactFacet::set_max_dimension ( )  [override], [virtual]
```

calculate the max dimension of the facet for range limit determination.

Implements jeod::ContactFacet.

Definition at line 202 of file line_contact_facet.cc.

References length, jeod::ContactFacet::max_dimension, and jeod::PointContactFacet::radius.

Referenced by jeod::LineContactFacetFactory::create_facet().

8.9.4 Field Documentation

8.9.4.1 length

```
double jeod::LineContactFacet::length {}
```

length of the line along the vehicle point x axis.

trick_units(m)

Definition at line 93 of file line_contact_facet.hh.

Referenced by calculate_contact_point(), jeod::LineContactFacetFactory::create_facet(), jeod::LineContactPair::in_contact(), jeod::LinePointContactPair::in_contact(), and set_max_dimension().

The documentation for this class was generated from the following files:

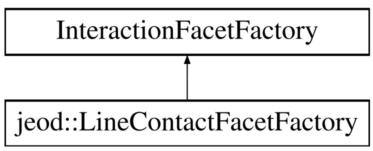
- line_contact_facet.hh
- line_contact_facet.cc

8.10 jeod::LineContactFacetFactory Class Reference

Creates a PointContactFacet from an InteractionFacet.

```
#include <line_contact_facet_factory.hh>
```

Inheritance diagram for jeod::LineContactFacetFactory:



Public Member Functions

- LineContactFacetFactory ()

  Default Constructor.
- ∼LineContactFacetFactory () override=default
- LineContactFacetFactory & operator= (const LineContactFacetFactory &)=delete
- LineContactFacetFactory (const LineContactFacetFactory &)=delete
- InteractionFacet ∗ create_facet (Facet ∗facet, FacetParams ∗params) override

  Create a PointContactFacet from a CircularFlatPlate facet and a ContactParams object.
- bool is_correct_factory (Facet ∗facet) override

  PointContactFacetFactory specific implementation of this function.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, LineContactFacetFactory)

8.10.1 Detailed Description

Creates a PointContactFacet from an InteractionFacet.

Definition at line 86 of file line_contact_facet_factory.hh.

8.10.2 Constructor & Destructor Documentation

8.10.2.1 LineContactFacetFactory() [1/2]

```
jeod::LineContactFacetFactory::LineContactFacetFactory ( )
```

Default Constructor.

Definition at line 52 of file line_contact_facet_factory.cc.

8.10.2.2 ∼LineContactFacetFactory()

```
jeod::LineContactFacetFactory::∼LineContactFacetFactory ( ) [override], [default]
```

8.10.2.3 LineContactFacetFactory() [2/2]

```
jeod::LineContactFacetFactory::LineContactFacetFactory (
            const LineContactFacetFactory & ) [delete]
```

8.10.3 Member Function Documentation

8.10.3.1 create_facet()

```
InteractionFacet * jeod::LineContactFacetFactory::create_facet (
            Facet * facet,
            FacetParams * params )  [override]
```

Create a PointContactFacet from a CircularFlatPlate facet and a ContactParams object.

Returns

The new EllipsoidContactFacet. Note that this is allocated and YOU are responsible for destroying it at the end!

Parameters

in	*facet*	The CircularFlatPlate. This MUST be a circular flat plate or the algorithm will send a failure message
in	*params*	ContactParams

Definition at line 65 of file line_contact_facet_factory.cc.

References jeod::ContactFacet::create_vehicle_point(), jeod::ContactMessages::initialization_error, jeod::Line↩ContactFacet::length, jeod::ContactFacet::normal, jeod::ContactFacet::position, jeod::PointContactFacet::radius, jeod::LineContactFacet::set_max_dimension(), jeod::ContactFacet::surface_type, and jeod::ContactFacet↩::vehicle_body.

8.10.3.2 is_correct_factory()

```
bool jeod::LineContactFacetFactory::is_correct_factory (
            Facet * facet )  [override]
```

PointContactFacetFactory specific implementation of this function.

If the Facet is of type CircularFlatPlate, returns true. False otherwise

Returns

true if facet is a FlatPlateCircular, false otherwise

Parameters

in	*facet*	The facet to check

Definition at line 135 of file line_contact_facet_factory.cc.

8.10.3.3 operator=()

```
LineContactFacetFactory& jeod::LineContactFacetFactory::operator= (
            const LineContactFacetFactory & ) [delete]
```

8.10.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::LineContactFacetFactory::p (
            jeod ,
            LineContactFacetFactory ) [private]
```

The documentation for this class was generated from the following files:

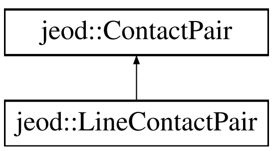
- line_contact_facet_factory.hh
- line_contact_facet_factory.cc

8.11 jeod::LineContactPair Class Reference

An point to point contact pair for use in the contact model.

```
#include <line_contact_pair.hh>
```

Inheritance diagram for jeod::LineContactPair:



Public Member Functions

- LineContactPair ()=default
- ∼LineContactPair () override=default
- LineContactPair & operator= (const LineContactPair &)=delete
- LineContactPair (const LineContactPair &)=delete
- void in_contact () override

    Determine if contact has occurred between the facets of the pair.
- void initialize_pair (ContactFacet ∗subject_facet, ContactFacet ∗target_facet) override

    Initialize the contact pair by setting the subject, target, and creating the relstate if possible.

Data Fields

- LineContactFacet ∗ line_subject {}

  pointer to the contact facet that is the subject of the associated relative states.
- LineContactFacet ∗ line_target {}

  pointer to the contact facet that is the target of the associated relative states.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, LineContactPair)

Additional Inherited Members

8.11.1 Detailed Description

An point to point contact pair for use in the contact model.

Definition at line 84 of file line_contact_pair.hh.

8.11.2 Constructor & Destructor Documentation

8.11.2.1 LineContactPair() [1/2]

```
jeod::LineContactPair::LineContactPair ( )  [default]
```

8.11.2.2 ∼LineContactPair()

```
jeod::LineContactPair::∼LineContactPair ( )  [override], [default]
```

8.11.2.3 LineContactPair() [2/2]

```
jeod::LineContactPair::LineContactPair (
            const LineContactPair & )  [delete]
```

8.11.3 Member Function Documentation

8.11.3.1 in_contact()

```
void jeod::LineContactPair::in_contact ( ) [override], [virtual]
```

Determine if contact has occurred between the facets of the pair.

Implements jeod::ContactPair.

Definition at line 48 of file line_contact_pair.cc.

References jeod::LineContactFacet::calculate_contact_point(), jeod::PairInteraction::calculate_forces(), jeod::↩
PointContactFacet::contact_point, jeod::ContactUtils::dist_line_segments(), jeod::ContactPair::interaction, jeod::↩
LineContactFacet::length, line_subject, line_target, and jeod::ContactPair::rel_state.

8.11.3.2 initialize_pair()

```
void jeod::LineContactPair::initialize_pair (
            ContactFacet * subject_facet,
            ContactFacet * target_facet ) [override], [virtual]
```

Initialize the contact pair by setting the subject, target, and creating the relstate if possible.

Parameters

in,out	*subject_facet*	subject ContactFacet
in,out	*target_facet*	target ContactFacet

Implements jeod::ContactPair.

Definition at line 138 of file line_contact_pair.cc.

References line_subject, line_target, jeod::ContactPair::rel_state, jeod::ContactPair::subject, jeod::ContactPair↩
::target, and jeod::ContactFacet::vehicle_point.

Referenced by jeod::LineContactFacet::create_pair().

8.11.3.3 operator=()

```
LineContactPair& jeod::LineContactPair::operator= (
            const LineContactPair & ) [delete]
```

8.11.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::LineContactPair::p (
            jeod ,
            LineContactPair ) [private]
```

8.11.4 Field Documentation

8.11.4.1 line_subject

`LineContactFacet* jeod::LineContactPair::line_subject {}`

pointer to the contact facet that is the subject of the associated relative states.

trick_units(--)

Definition at line 90 of file line_contact_pair.hh.

Referenced by in_contact(), and initialize_pair().

8.11.4.2 line_target

`LineContactFacet* jeod::LineContactPair::line_target {}`

pointer to the contact facet that is the target of the associated relative states.

trick_units(--)

Definition at line 95 of file line_contact_pair.hh.

Referenced by in_contact(), and initialize_pair().

The documentation for this class was generated from the following files:

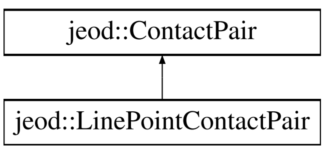
- line_contact_pair.hh
- line_contact_pair.cc

8.12 jeod::LinePointContactPair Class Reference

An point to point contact pair for use in the contact model.

`#include <line_point_contact_pair.hh>`

Inheritance diagram for jeod::LinePointContactPair:



Public Member Functions

- LinePointContactPair ()=default
- ∼LinePointContactPair () override=default
- LinePointContactPair & operator= (const LineContactPair &)=delete
- LinePointContactPair (const LineContactPair &)=delete
- void in_contact () override

  Determine if contact has occurred between the facets of the pair.
- void initialize_pair (ContactFacet ∗subject_facet, ContactFacet ∗target_facet) override

  Initialize the contact pair by setting the subject, target, and creating the relstate if possible.

Data Fields

- LineContactFacet ∗ line_subject {}

  pointer to the contact facet that is the subject of the associated relative states.
- PointContactFacet ∗ point_target {}

  pointer to the contact facet that is the target of the associated relative states.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, LinePointContactPair)

Additional Inherited Members

8.12.1 Detailed Description

An point to point contact pair for use in the contact model.

Definition at line 85 of file line_point_contact_pair.hh.

8.12.2 Constructor & Destructor Documentation

8.12.2.1 LinePointContactPair() [1/2]

```
jeod::LinePointContactPair::LinePointContactPair ( )  [default]
```

8.12.2.2 ∼LinePointContactPair()

```
jeod::LinePointContactPair::∼LinePointContactPair ( )  [override], [default]
```

8.12.2.3 LinePointContactPair() [2/2]

```
jeod::LinePointContactPair::LinePointContactPair (
            const LineContactPair & ) [delete]
```

8.12.3 Member Function Documentation

8.12.3.1 in_contact()

```
void jeod::LinePointContactPair::in_contact ( ) [override], [virtual]
```

Determine if contact has occurred between the facets of the pair.

Implements jeod::ContactPair.

Definition at line 49 of file line_point_contact_pair.cc.

References jeod::PointContactFacet::calculate_contact_point(), jeod::LineContactFacet::calculate_contact_point(), jeod::PairInteraction::calculate_forces(), jeod::PointContactFacet::contact_point, jeod::ContactUtils::dist_line_↩ segments(), jeod::ContactPair::interaction, jeod::LineContactFacet::length, line_subject, point_target, and jeod::↩ ContactPair::rel_state.

8.12.3.2 initialize_pair()

```
void jeod::LinePointContactPair::initialize_pair (
            ContactFacet * subject_facet,
            ContactFacet * target_facet ) [override], [virtual]
```

Initialize the contact pair by setting the subject, target, and creating the relstate if possible.

Parameters

in,out	*subject_facet*	subject ContactFacet
in,out	*target_facet*	target ContactFacet

Implements jeod::ContactPair.

Definition at line 132 of file line_point_contact_pair.cc.

References line_subject, point_target, jeod::ContactPair::rel_state, jeod::ContactPair::subject, jeod::ContactPair↩ ::target, and jeod::ContactFacet::vehicle_point.

Referenced by jeod::LineContactFacet::create_pair().

8.12.3.3 operator=()

```
LinePointContactPair& jeod::LinePointContactPair::operator= (
            const LineContactPair & ) [delete]
```

8.12.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::LinePointContactPair::p (
            jeod ,
            LinePointContactPair ) [private]
```

8.12.4 Field Documentation

8.12.4.1 line_subject

```
LineContactFacet* jeod::LinePointContactPair::line_subject {}
```

pointer to the contact facet that is the subject of the associated relative states.

trick_units(–)

Definition at line 91 of file line_point_contact_pair.hh.

Referenced by in_contact(), and initialize_pair().

8.12.4.2 point_target

```
PointContactFacet* jeod::LinePointContactPair::point_target {}
```

pointer to the contact facet that is the target of the associated relative states.

trick_units(–)

Definition at line 96 of file line_point_contact_pair.hh.

Referenced by in_contact(), and initialize_pair().

The documentation for this class was generated from the following files:

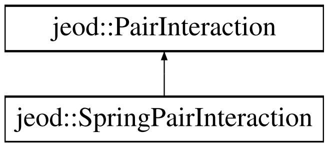
- line_point_contact_pair.hh
- line_point_contact_pair.cc

8.13 jeod::PairInteraction Class Reference

Simple spring contact parameters.

```
#include <pair_interaction.hh>
```

Inheritance diagram for jeod::PairInteraction:



Public Member Functions

- PairInteraction ()=default
- virtual ∼PairInteraction ()=default
- PairInteraction & operator= (const PairInteraction &)=delete
- PairInteraction (const PairInteraction &)=delete
- bool is_correct_interaction (ContactParams ∗subject_params, ContactParams ∗target_params)

    Check a pair of contact params for a match to the ones defined for this pair_interaction.
- virtual void calculate_forces (ContactFacet ∗subject, ContactFacet ∗target, RelativeDerivedState ∗rel_state, double ∗penetration_vector, double ∗rel_velocity)=0

    Pure virtual function that is defined to calculate forces on facets in contact.

Data Fields

- std::string params_1

    contact param type that defines this pair interaction.
- std::string params_2

    contact param type that defines this pair interaction.
- double friction_mag {}

    magnitude of the friction force on the contact surfaces.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, PairInteraction)

8.13.1 Detailed Description

Simple spring contact parameters.

Definition at line 89 of file pair_interaction.hh.

8.13.2 Constructor & Destructor Documentation

8.13.2.1 PairInteraction() [1/2]

```
jeod::PairInteraction::PairInteraction ( ) [default]
```

8.13.2.2 ∼PairInteraction()

```
virtual jeod::PairInteraction::~PairInteraction ( ) [virtual], [default]
```

8.13.2.3 PairInteraction() [2/2]

```
jeod::PairInteraction::PairInteraction (
            const PairInteraction & ) [delete]
```

8.13.3 Member Function Documentation

8.13.3.1 calculate_forces()

```
virtual void jeod::PairInteraction::calculate_forces (
            ContactFacet * subject,
            ContactFacet * target,
            RelativeDerivedState * rel_state,
            double * penetration_vector,
            double * rel_velocity ) [pure virtual]
```

Pure virtual function that is defined to calculate forces on facets in contact.

Parameters

in,out	*subject*	subject of the relative state
in,out	*target*	target of the relative state
in	*rel_state*	relative state between subject and target in subject frame
in	*penetration_vector*	vector that characterises the interpenetration of the subject and the target
in	*rel_velocity*	relative velocity of the subject and the target in the subject frame

Implemented in jeod::SpringPairInteraction.

Referenced by jeod::LineContactPair::in_contact(), jeod::LinePointContactPair::in_contact(), and jeod::Point←ContactPair::in_contact().

8.13.3.2 is_correct_interaction()

```
bool jeod::PairInteraction::is_correct_interaction (
            ContactParams * subject_params,
            ContactParams * target_params )
```

Check a pair of contact params for a match to the ones defined for this pair_interaction.

Returns

bool

Parameters

in	*subject_params*	parameters of the subject
in	*target_params*	parameters of the target

Definition at line 50 of file pair_interaction.cc.

References params_1, and params_2.

8.13.3.3 operator=()

```
PairInteraction& jeod::PairInteraction::operator= (
            const PairInteraction & )  [delete]
```

8.13.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::PairInteraction::p (
            jeod ,
            PairInteraction )  [private]
```

8.13.4 Field Documentation

8.13.4.1 friction_mag

```
double jeod::PairInteraction::friction_mag {}
```

magnitude of the friction force on the contact surfaces.

trick_units(N)

Definition at line 105 of file pair_interaction.hh.

Referenced by jeod::SpringPairInteraction::calculate_forces().

8.13.4.2 params_1

`std::string jeod::PairInteraction::params_1`

contact param type that defines this pair interaction.

trick_units(--)

Definition at line 95 of file pair_interaction.hh.

Referenced by is_correct_interaction().

8.13.4.3 params_2

`std::string jeod::PairInteraction::params_2`

contact param type that defines this pair interaction.

trick_units(--)

Definition at line 100 of file pair_interaction.hh.

Referenced by is_correct_interaction().

The documentation for this class was generated from the following files:

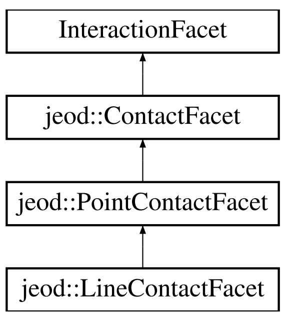
- pair_interaction.hh
- pair_interaction.cc

8.14 jeod::PointContactFacet Class Reference

The contact facet based on the distance to a single point, specifically the vehicle point.

`#include <point_contact_facet.hh>`

Inheritance diagram for jeod::PointContactFacet:



Public Member Functions

- PointContactFacet ()

    Default constructor.
- ∼PointContactFacet () override=default
- PointContactFacet & operator= (const PointContactFacet &)=delete
- PointContactFacet (const PointContactFacet &)=delete
- ContactPair ∗ create_pair () override

    Overloaded functions that create a ContactPair and pass the address of it to the Contact class for addition to the list of pairs.
- ContactPair ∗ create_pair (ContactFacet ∗target, Contact ∗contact) override

    Overloaded functions that create a ContactPair and pass the address of it to the Contact class for addition to the list of pairs.
- void set_max_dimension () override

    calculate the max dimension of the facet for range limit determination.
- virtual void calculate_contact_point (double nvec[3])

    Use the relstate and radius of contact to calculate a contact point on this facet.
- void calculate_torque (double ∗tmp_force) override

    Calculate the torque generated on the vehicle by the facet.

Data Fields

- double radius {}

    radius from the point at which contact takes place.
- double contact_point [3] {}

    Contact point given in facet vehicle point frame, representing point on the point on the surface of a sphere of radius "radius" where contact has occured.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, PointContactFacet)

8.14.1 Detailed Description

The contact facet based on the distance to a single point, specifically the vehicle point.

In effect this represents a sphere.

Definition at line 87 of file point_contact_facet.hh.

8.14.2 Constructor & Destructor Documentation

8.14.2.1 PointContactFacet() [1/2]

```
jeod::PointContactFacet::PointContactFacet ( )
```

Default constructor.

Definition at line 50 of file point_contact_facet.cc.

8.14.2.2 ~PointContactFacet()

```
jeod::PointContactFacet::~PointContactFacet ( )  [override], [default]
```

8.14.2.3 PointContactFacet() [2/2]

```
jeod::PointContactFacet::PointContactFacet (
            const PointContactFacet & )  [delete]
```

8.14.3 Member Function Documentation

8.14.3.1 calculate_contact_point()

```
void jeod::PointContactFacet::calculate_contact_point (
            double nvec[3] )  [virtual]
```

Use the relstate and radius of contact to calculate a contact point on this facet.

Parameters

in	*nvec*	direction vector between the two facets

Reimplemented in jeod::LineContactFacet.

Definition at line 119 of file point_contact_facet.cc.

References contact_point, and radius.

Referenced by jeod::LinePointContactPair::in_contact(), and jeod::PointContactPair::in_contact().

8.14.3.2 calculate_torque()

```
void jeod::PointContactFacet::calculate_torque (
            double * tmp_force )  [override], [virtual]
```

Calculate the torque generated on the vehicle by the facet.

Assumes that the force is in the vehicle structural frame, but that close_point is not.

Parameters

in	*tmp_force*	force from one contact interaction. Units: N

Implements jeod::ContactFacet.

Definition at line 141 of file point_contact_facet.cc.

References contact_point, jeod::ContactFacet::vehicle_body, and jeod::ContactFacet::vehicle_point.

8.14.3.3 create_pair() [1/2]

```
ContactPair * jeod::PointContactFacet::create_pair ( )  [override], [virtual]
```

Overloaded functions that create a ContactPair and pass the address of it to the Contact class for addition to the list of pairs.

This function is called to create a pair that only contains a subject.

Returns

ContactPair that was created

Implements jeod::ContactFacet.

Definition at line 61 of file point_contact_facet.cc.

References jeod::PointContactPair::initialize_pair().

8.14.3.4 create_pair() [2/2]

```
ContactPair * jeod::PointContactFacet::create_pair (
            ContactFacet * target,
            Contact * contact )  [override], [virtual]
```

Overloaded functions that create a ContactPair and pass the address of it to the Contact class for addition to the list of pairs.

This function is called when a subject and target are known.

Returns

ContactPair that was created

Parameters

in,out	*target*	target ContactFacet
in	*contact*	Contact object used to find the pair interaction

Implements jeod::ContactFacet.

Definition at line 80 of file point_contact_facet.cc.

References jeod::Contact::contact_limit_factor, jeod::Contact::find_interaction(), jeod::ContactMessages↩ ::initialization_warns, jeod::PointContactPair::initialize_pair(), jeod::ContactPair::interaction, jeod::ContactPair↩ ::interaction_distance, jeod::ContactFacet::max_dimension, and jeod::ContactFacet::surface_type.

8.14.3.5 operator=()

```
PointContactFacet& jeod::PointContactFacet::operator= (
            const PointContactFacet & )  [delete]
```

8.14.3.6 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::PointContactFacet::p (
            jeod ,
            PointContactFacet  )  [private]
```

8.14.3.7 set_max_dimension()

```
void jeod::PointContactFacet::set_max_dimension ( )  [override], [virtual]
```

calculate the max dimension of the facet for range limit determination.

Implements jeod::ContactFacet.

Definition at line 131 of file point_contact_facet.cc.

References jeod::ContactFacet::max_dimension, and radius.

Referenced by jeod::PointContactFacetFactory::create_facet().

8.14.4 Field Documentation

8.14.4.1 contact_point

```
double jeod::PointContactFacet::contact_point[3] {}
```

Contact point given in facet vehicle point frame, representing point on the point on the surface of a sphere of radius "radius" where contact has occured.

trick_units(m)

Definition at line 100 of file point_contact_facet.hh.

Referenced by calculate_contact_point(), jeod::LineContactFacet::calculate_contact_point(), jeod::LineContactFacet::calculate_torque(), calculate_torque(), jeod::LineContactPair::in_contact(), jeod::LinePointContactPair::in_contact(), and jeod::PointContactPair::in_contact().

8.14.4.2 radius

```
double jeod::PointContactFacet::radius {}
```

radius from the point at which contact takes place.

trick_units(m)

Definition at line 93 of file point_contact_facet.hh.

Referenced by calculate_contact_point(), jeod::LineContactFacet::calculate_contact_point(), jeod::LineContactFacetFactory::create_facet(), jeod::PointContactFacetFactory::create_facet(), jeod::PointContactPair::in_contact(), jeod::LineContactFacet::set_max_dimension(), and set_max_dimension().

The documentation for this class was generated from the following files:

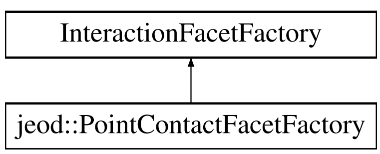
- point_contact_facet.hh
- point_contact_facet.cc

8.15 jeod::PointContactFacetFactory Class Reference

Creates a PointContactFacet from an InteractionFacet.

```
#include <point_contact_facet_factory.hh>
```

Inheritance diagram for jeod::PointContactFacetFactory:



Public Member Functions

- PointContactFacetFactory ()
  Default Constructor.
- ∼PointContactFacetFactory () override=default
- PointContactFacetFactory & operator= (const PointContactFacetFactory &)=delete
- PointContactFacetFactory (const PointContactFacetFactory &)=delete
- InteractionFacet ∗ create_facet (Facet ∗facet, FacetParams ∗params) override
  Create a PointContactFacet from a CircularFlatPlate facet and a ContactParams object.
- bool is_correct_factory (Facet ∗facet) override
  PointContactFacetFactory specific implementation of this function.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, PointContactFacetFactory)

8.15.1 Detailed Description

Creates a PointContactFacet from an InteractionFacet.

Definition at line 86 of file point_contact_facet_factory.hh.

8.15.2 Constructor & Destructor Documentation

8.15.2.1 PointContactFacetFactory() [1/2]

```
jeod::PointContactFacetFactory::PointContactFacetFactory ( )
```

Default Constructor.

Definition at line 51 of file point_contact_facet_factory.cc.

8.15.2.2 ∼PointContactFacetFactory()

```
jeod::PointContactFacetFactory::∼PointContactFacetFactory ( ) [override], [default]
```

8.15.2.3 PointContactFacetFactory() [2/2]

```
jeod::PointContactFacetFactory::PointContactFacetFactory (
            const PointContactFacetFactory & ) [delete]
```

8.15.3 Member Function Documentation

8.15.3.1 create_facet()

```
InteractionFacet * jeod::PointContactFacetFactory::create_facet (
            Facet * facet,
            FacetParams * params )  [override]
```

Create a PointContactFacet from a CircularFlatPlate facet and a ContactParams object.

Returns

The new EllipsoidContactFacet. Note that this is allocated and YOU are responsible for destroying it at the end!

Parameters

in	*facet*	The CircularFlatPlate. This MUST be a circular flat plate or the algorithm will send a failure message
in	*params*	ContactParams

Definition at line 64 of file point_contact_facet_factory.cc.

References jeod::ContactFacet::create_vehicle_point(), jeod::ContactMessages::initialization_error, jeod::ContactFacet::normal, jeod::ContactFacet::position, jeod::PointContactFacet::radius, jeod::PointContactFacet::set_max_dimension(), jeod::ContactFacet::surface_type, and jeod::ContactFacet::vehicle_body.

8.15.3.2 is_correct_factory()

```
bool jeod::PointContactFacetFactory::is_correct_factory (
            Facet * facet )  [override]
```

PointContactFacetFactory specific implementation of this function.

If the Facet is of type CircularFlatPlate, returns true. False otherwise

Returns

true if facet is a FlatPlateCircular, false otherwise

Parameters

in	*facet*	The facet to check

Definition at line 133 of file point_contact_facet_factory.cc.

8.15.3.3 operator=()

```
PointContactFacetFactory& jeod::PointContactFacetFactory::operator= (
            const PointContactFacetFactory & )  [delete]
```

8.15.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::PointContactFacetFactory::p (
            jeod ,
            PointContactFacetFactory )  [private]
```

The documentation for this class was generated from the following files:

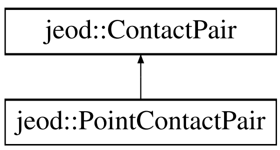
- point_contact_facet_factory.hh
- point_contact_facet_factory.cc

8.16 jeod::PointContactPair Class Reference

An point to point contact pair for use in the contact model.

```
#include <point_contact_pair.hh>
```

Inheritance diagram for jeod::PointContactPair:



Public Member Functions

- PointContactPair ()=default
- ∼PointContactPair () override=default
- PointContactPair & operator= (const PointContactPair &)=delete
- PointContactPair (const PointContactPair &)=delete
- void in_contact () override

  Determine if contact has occurred between the facets of the pair.
- void initialize_pair (ContactFacet ∗subject_facet, ContactFacet ∗target_facet) override

  Initialize the contact pair by setting the subject, target, and creating the relstate if possible.

Data Fields

- PointContactFacet ∗ point_subject {}

  pointer to the point contact facet that is the subject of the associated relative states.
- PointContactFacet ∗ point_target {}

  pointer to the point contact facet that is the target of the associated relative states.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, PointContactPair)

Additional Inherited Members

8.16.1 Detailed Description

An point to point contact pair for use in the contact model.

Definition at line 84 of file point_contact_pair.hh.

8.16.2 Constructor & Destructor Documentation

8.16.2.1 PointContactPair() [1/2]

```
jeod::PointContactPair::PointContactPair ( ) [default]
```

8.16.2.2 ∼PointContactPair()

```
jeod::PointContactPair::∼PointContactPair ( ) [override], [default]
```

8.16.2.3 PointContactPair() [2/2]

```
jeod::PointContactPair::PointContactPair (
            const PointContactPair & ) [delete]
```

8.16.3 Member Function Documentation

8.16.3.1 in_contact()

```
void jeod::PointContactPair::in_contact ( )  [override], [virtual]
```

Determine if contact has occurred between the facets of the pair.

Implements jeod::ContactPair.

Definition at line 47 of file point_contact_pair.cc.

References jeod::PointContactFacet::calculate_contact_point(), jeod::PairInteraction::calculate_forces(), jeod::↩ PointContactFacet::contact_point, jeod::ContactPair::interaction, point_subject, point_target, jeod::PointContact↩ Facet::radius, and jeod::ContactPair::rel_state.

8.16.3.2 initialize_pair()

```
void jeod::PointContactPair::initialize_pair (
            ContactFacet * subject_facet,
            ContactFacet * target_facet )  [override], [virtual]
```

Initialize the contact pair by setting the subject, target, and creating the relstate if possible.

Parameters

in,out	*subject_facet*	subject ContactFacet
in,out	*target_facet*	target ContactFacet

Implements jeod::ContactPair.

Definition at line 96 of file point_contact_pair.cc.

References point_subject, point_target, jeod::ContactPair::rel_state, jeod::ContactPair::subject, jeod::ContactPair↩ ::target, and jeod::ContactFacet::vehicle_point.

Referenced by jeod::PointContactFacet::create_pair().

8.16.3.3 operator=()

```
PointContactPair& jeod::PointContactPair::operator= (
            const PointContactPair &  )  [delete]
```

8.16.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::PointContactPair::p (
            jeod ,
            PointContactPair  )  [private]
```

8.16.4 Field Documentation

8.16.4.1 point_subject

`PointContactFacet* jeod::PointContactPair::point_subject {}`

pointer to the point contact facet that is the subject of the associated relative states.

trick_units(--)

Definition at line 90 of file point_contact_pair.hh.

Referenced by in_contact(), and initialize_pair().

8.16.4.2 point_target

`PointContactFacet* jeod::PointContactPair::point_target {}`

pointer to the point contact facet that is the target of the associated relative states.

trick_units(--)

Definition at line 95 of file point_contact_pair.hh.

Referenced by in_contact(), and initialize_pair().

The documentation for this class was generated from the following files:

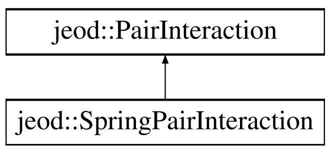
- point_contact_pair.hh
- point_contact_pair.cc

8.17 jeod::SpringPairInteraction Class Reference

Simple spring contact parameters.

`#include <spring_pair_interaction.hh>`

Inheritance diagram for jeod::SpringPairInteraction:



Public Member Functions

- SpringPairInteraction ()=default
- ∼SpringPairInteraction () override=default
- SpringPairInteraction & operator= (const SpringPairInteraction &)=delete
- SpringPairInteraction (const SpringPairInteraction &)=delete
- void calculate_forces (ContactFacet ∗subject, ContactFacet ∗target, RelativeDerivedState ∗rel_state, double ∗penetration_vector, double ∗rel_velocity) override

  force calculation for a simple spring based contact dynamics model, takes in geometry information from the appropriate ContactFacet::calculate_forces but doesn't know about specific type of ContactFacet

Data Fields

- double spring_k {}

  Spring stiffness constant.
- double damping_b {}

  Spring damping constant.
- double mu {}

  Coefficent of friction.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, SpringPairInteraction)

8.17.1 Detailed Description

Simple spring contact parameters.

Definition at line 83 of file spring_pair_interaction.hh.

8.17.2 Constructor & Destructor Documentation

8.17.2.1 SpringPairInteraction() [1/2]

```
jeod::SpringPairInteraction::SpringPairInteraction ( ) [default]
```

8.17.2.2 ∼SpringPairInteraction()

```
jeod::SpringPairInteraction::~SpringPairInteraction ( ) [override], [default]
```

8.17.2.3 SpringPairInteraction() [2/2]

```
jeod::SpringPairInteraction::SpringPairInteraction (
            const SpringPairInteraction & )  [delete]
```

8.17.3 Member Function Documentation

8.17.3.1 calculate_forces()

```
void jeod::SpringPairInteraction::calculate_forces (
            ContactFacet * subject,
            ContactFacet * target,
            RelativeDerivedState * rel_state,
            double * penetration_vector,
            double * rel_velocity )  [override], [virtual]
```

force calculation for a simple spring based contact dynamics model, takes in geometry information from the appropriate ContactFacet::calculate_forces but doesn't know about specific type of ContactFacet

Parameters

in,out	*subject*	subject frame of the relative state
in,out	*target*	target frame of the relative state
in	*rel_state*	relative state between subject and target in subject frame
in	*penetration_vector*	vector that characterises the interpenetration of the subject and the target
in	*rel_velocity*	relative velocity of the subject and the target in the subject frame

Implements jeod::PairInteraction.

Definition at line 57 of file spring_pair_interaction.cc.

References jeod::ContactFacet::calculate_torque(), damping_b, jeod::PairInteraction::friction_mag, mu, spring_k, jeod::ContactFacet::vehicle_body, and jeod::ContactFacet::vehicle_point.

8.17.3.2 operator=()

```
SpringPairInteraction& jeod::SpringPairInteraction::operator= (
            const SpringPairInteraction & )  [delete]
```

8.17.3.3 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::SpringPairInteraction::p (
            jeod ,
            SpringPairInteraction )  [private]
```

8.17.4 Field Documentation

8.17.4.1 damping_b

```
double jeod::SpringPairInteraction::damping_b {}
```

Spring damping constant.

trick_units(N∗s/m)

Definition at line 94 of file spring_pair_interaction.hh.

Referenced by calculate_forces().

8.17.4.2 mu

```
double jeod::SpringPairInteraction::mu {}
```

Coefficent of friction.

trick_units(–)

Definition at line 99 of file spring_pair_interaction.hh.

Referenced by calculate_forces().

8.17.4.3 spring_k

```
double jeod::SpringPairInteraction::spring_k {}
```

Spring stiffness constant.

trick_units(N/m)

Definition at line 89 of file spring_pair_interaction.hh.

Referenced by calculate_forces().

The documentation for this class was generated from the following files:

- spring_pair_interaction.hh
- spring_pair_interaction.cc

Chapter 9

File Documentation

9.1 class_declarations.hh File Reference

Forward declaration of classes defined in the contact model.

Namespaces

- jeod

    Namespace jeod.

9.1.1 Detailed Description

Forward declaration of classes defined in the contact model.

9.2 contact.cc File Reference

Base Contact for use with contact interaction model.

```
#include "dynamics/mass/include/mass.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/surface_model/include/facet.hh"
#include "../include/contact.hh"
#include "../include/contact_facet.hh"
#include "../include/contact_pair.hh"
#include "../include/pair_interaction.hh"
```

Namespaces

- jeod

    Namespace jeod.

9.2.1 Detailed Description

Base Contact for use with contact interaction model.

9.3 contact.hh File Reference

(Base class to for the contact manager for use with contact interaction model)

```
#include <list>
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "utils/container/include/pointer_list.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "contact_facet.hh"
#include "contact_pair.hh"
#include "pair_interaction.hh"
```

Data Structures

- class jeod::Contact

  An base contact class for use in the surface model.

Namespaces

- jeod

  Namespace jeod.

9.3.1 Detailed Description

(Base class to for the contact manager for use with contact interaction model)

9.4 contact_facet.cc File Reference

Define ContactFacet::create_vehicle_point.

```
#include <cstddef>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/mass/include/mass_point_init.hh"
#include "utils/math/include/numerical.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/surface_model/include/facet.hh"
#include "../include/contact_facet.hh"
#include "../include/contact_messages.hh"
#include "../include/contact_utils.hh"
```

Namespaces

- jeod

    Namespace jeod.

9.4.1 Detailed Description

Define ContactFacet::create_vehicle_point.

9.5 contact_facet.hh File Reference

Individual facets for use with contact interaction models.

```
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/surface_model/include/facet.hh"
#include "utils/surface_model/include/interaction_facet.hh"
#include "class_declarations.hh"
```

Data Structures

- class jeod::ContactFacet

    An contact interaction specific facet for use in the surface model.

Namespaces

- jeod

    Namespace jeod.

9.5.1 Detailed Description

Individual facets for use with contact interaction models.

9.6 contact_messages.cc File Reference

Implement contact_messages.

```
#include "utils/message/include/make_message_code.hh"
#include "../include/contact_messages.hh"
```

Namespaces

- jeod

    Namespace jeod.

Macros

- #define MAKE_CONTACT_MESSAGE_CODE(id) JEOD_MAKE_MESSAGE_CODE(ContactMessages, "interactions/contact", id)

9.6.1 Detailed Description

Implement contact_messages.

9.6.2 Macro Definition Documentation

9.6.2.1 MAKE_CONTACT_MESSAGE_CODE

```
#define MAKE_CONTACT_MESSAGE_CODE(
              id ) JEOD_MAKE_MESSAGE_CODE(ContactMessages, "interactions/contact", id)
```

Definition at line 43 of file contact_messages.cc.

9.7 contact_messages.hh File Reference

Contact message for message handling.

```
#include "utils/message/include/message_handler.hh"
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class jeod::ContactMessages

    Messages associated with use of the contact model.

Namespaces

- jeod

    Namespace jeod.

9.7.1 Detailed Description

Contact message for message handling.

9.8 contact_pair.cc File Reference

ContactPair class for use with contact interaction model.

```
#include <cmath>
#include "dynamics/mass/include/mass.hh"
#include "utils/math/include/vector3.hh"
#include "../include/contact_pair.hh"
```

Namespaces

- jeod

  Namespace jeod.

9.8.1 Detailed Description

ContactPair class for use with contact interaction model.

9.9 contact_pair.hh File Reference

Base class for pair of contact facets for use with contact interaction model.

```
#include "dynamics/derived_state/include/relative_derived_state.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "contact_facet.hh"
```

Data Structures

- class jeod::ContactPair

  An base contact pair class for use in the contact model.

Namespaces

- jeod

  Namespace jeod.

9.9.1 Detailed Description

Base class for pair of contact facets for use with contact interaction model.

9.10 contact_params.cc File Reference

contact parameters for use in the surface model

```
#include "utils/memory/include/jeod_alloc.hh"
#include "../include/contact_params.hh"
```

Namespaces

- jeod

  Namespace jeod.

9.10.1 Detailed Description

contact parameters for use in the surface model

9.11 contact_params.hh File Reference

A class for contact facet parameters, used to create interaction facets for contact in the InteractionSurfaceFactorys.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/surface_model/include/facet_params.hh"
```

Data Structures

- class jeod::ContactParams

  A base class for all contact parameters used in the surface model.

Namespaces

- jeod

  Namespace jeod.

9.11.1 Detailed Description

A class for contact facet parameters, used to create interaction facets for contact in the InteractionSurfaceFactorys.

9.12 contact_surface.cc File Reference

Vehicle surface model for the contact interaction models.

```
#include <cstddef>
#include <typeinfo>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/surface_model/include/facet.hh"
#include "utils/surface_model/include/interaction_facet_factory.hh"
#include "../include/contact_facet.hh"
#include "../include/contact_messages.hh"
#include "../include/contact_surface.hh"
```

Namespaces

- jeod

    Namespace jeod.

9.12.1 Detailed Description

Vehicle surface model for the contact interaction models.

9.13 contact_surface.hh File Reference

Vehicle surface model for contact.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/surface_model/include/class_declarations.hh"
#include "utils/surface_model/include/interaction_surface.hh"
#include "contact_facet.hh"
```

Data Structures

- class jeod::ContactSurface

    The contact specific interaction surface, for use with the surface model.

Namespaces

- jeod

    Namespace jeod.

9.13.1 Detailed Description

Vehicle surface model for contact.

9.14 contact_surface_factory.cc File Reference

Factory that creates an contact surface, from a surface model.

```
#include <cstddef>
#include <cstring>
#include "dynamics/mass/include/mass.hh"
#include "utils/surface_model/include/facet.hh"
#include "utils/surface_model/include/facet_params.hh"
#include "utils/surface_model/include/interaction_surface.hh"
#include "utils/surface_model/include/surface_model.hh"
#include "../include/contact_messages.hh"
#include "../include/contact_params.hh"
#include "../include/contact_surface_factory.hh"
```

Namespaces

- jeod

  Namespace jeod.

9.14.1 Detailed Description

Factory that creates an contact surface, from a surface model.

9.15 contact_surface_factory.hh File Reference

Factory that creates an contact interaction surface from a surface model.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/surface_model/include/class_declarations.hh"
#include "utils/surface_model/include/interaction_surface_factory.hh"
#include "line_contact_facet_factory.hh"
#include "point_contact_facet_factory.hh"
```

Data Structures

- class jeod::ContactSurfaceFactory

  The surface factory that creates an contact specific surface from a general surface.

Namespaces

- jeod

    Namespace jeod.

9.15.1 Detailed Description

Factory that creates an contact interaction surface from a surface model.

9.16 contact_utils.hh File Reference

This Model is used for utility rotines.

```
#include "contact_utils_inline.hh"
```

Data Structures

- class jeod::ContactUtils

    Utility string and math functions for the contact model.

Namespaces

- jeod

    Namespace jeod.

9.16.1 Detailed Description

This Model is used for utility rotines.

9.17 contact_utils_inline.hh File Reference

Define ContactUtils::create_relstate_name, ContactUtils::copy_const_char_to_char.

```
#include <cstring>
#include "utils/math/include/vector3.hh"
#include "contact_messages.hh"
#include "contact_utils.hh"
```

Namespaces

- jeod

    Namespace jeod.

9.17.1 Detailed Description

Define ContactUtils::create_relstate_name, ContactUtils::copy_const_char_to_char.

9.18 line_contact_facet.cc File Reference

Define LineContactFacet functions.

```
#include <cmath>
#include <cstring>
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/numerical.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "../include/contact.hh"
#include "../include/contact_messages.hh"
#include "../include/contact_params.hh"
#include "../include/contact_utils.hh"
#include "../include/line_contact_facet.hh"
#include "../include/line_contact_pair.hh"
#include "../include/line_point_contact_pair.hh"
```

Namespaces

- jeod

    Namespace jeod.

9.18.1 Detailed Description

Define LineContactFacet functions.

9.19 line_contact_facet.hh File Reference

The contact facet based on the distance to a line segment centered on the vehicle point.

```
#include "dynamics/derived_state/include/relative_derived_state.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "point_contact_facet.hh"
#include "line_contact_pair.hh"
#include "line_point_contact_pair.hh"
```

Data Structures

- class jeod::LineContactFacet

    The contact facet based on the distance to a single point, specifically the vehicle point.

Namespaces

- jeod

    Namespace jeod.

9.19.1 Detailed Description

The contact facet based on the distance to a line segment centered on the vehicle point.

In effect this represents a cylinder with spherical ends.

9.20 line_contact_facet_factory.cc File Reference

Factory that creates a LineContactFacetFactory from a Cylinder facet and a ContactParams object.

```
#include <cstddef>
#include <typeinfo>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/surface_model/include/cylinder.hh"
#include "../include/contact_messages.hh"
#include "../include/contact_params.hh"
#include "../include/line_contact_facet.hh"
#include "../include/line_contact_facet_factory.hh"
```

Namespaces

- jeod

    Namespace jeod.

9.20.1 Detailed Description

Factory that creates a LineContactFacetFactory from a Cylinder facet and a ContactParams object.

9.21 line_contact_facet_factory.hh File Reference

Creates a line contact facet from an cylinder facet.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/surface_model/include/class_declarations.hh"
#include "utils/surface_model/include/interaction_facet.hh"
#include "utils/surface_model/include/interaction_facet_factory.hh"
#include "class_declarations.hh"
#include "line_contact_facet.hh"
```

Data Structures

- class jeod::LineContactFacetFactory

    Creates a PointContactFacet from an InteractionFacet.

Namespaces

- jeod

    Namespace jeod.

9.21.1 Detailed Description

Creates a line contact facet from an cylinder facet.

9.22 line_contact_pair.cc File Reference

LineContactPair class for use with contact interaction model.

```
#include "dynamics/mass/include/mass.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/contact_utils.hh"
#include "../include/line_contact_facet.hh"
#include "../include/line_contact_pair.hh"
#include "../include/pair_interaction.hh"
```

Namespaces

- jeod

    Namespace jeod.

9.22.1 Detailed Description

LineContactPair class for use with contact interaction model.

9.23 line_contact_pair.hh File Reference

Class for a pair of line contact facets for use with contact interaction model.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "contact_pair.hh"
```

Data Structures

- class jeod::LineContactPair

  An point to point contact pair for use in the contact model.

Namespaces

- jeod

  Namespace jeod.

9.23.1 Detailed Description

Class for a pair of line contact facets for use with contact interaction model.

9.24 line_point_contact_pair.cc File Reference

LinePointContactPair class for use with contact interaction model.

```
#include "dynamics/mass/include/mass.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/contact_utils.hh"
#include "../include/line_contact_facet.hh"
#include "../include/line_point_contact_pair.hh"
#include "../include/pair_interaction.hh"
#include "../include/point_contact_facet.hh"
```

Namespaces

- jeod

  Namespace jeod.

9.24.1 Detailed Description

LinePointContactPair class for use with contact interaction model.

9.25 line_point_contact_pair.hh File Reference

Class for a pair of a line contact facet and a point contact facet for use with contact interaction model.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "contact_pair.hh"
```

Data Structures

- class jeod::LinePointContactPair

    An point to point contact pair for use in the contact model.

Namespaces

- jeod

    Namespace jeod.

9.25.1 Detailed Description

Class for a pair of a line contact facet and a point contact facet for use with contact interaction model.

9.26 pair_interaction.cc File Reference

A class to define the interaction type for a pair of contact facets.

```
#include <cstring>
#include "../include/contact_facet.hh"
#include "../include/contact_params.hh"
#include "../include/pair_interaction.hh"
```

Namespaces

- jeod

    Namespace jeod.

9.26.1 Detailed Description

A class to define the interaction type for a pair of contact facets.

This is a base class and derived classes define the force generation function when contact between facets occurs.

9.27 pair_interaction.hh File Reference

A class to define the interaction type for a pair of contact facets.

```
#include <string>
#include "dynamics/derived_state/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "../include/class_declarations.hh"
```

Data Structures

- class jeod::PairInteraction

  Simple spring contact parameters.

Namespaces

- jeod

  Namespace jeod.

9.27.1 Detailed Description

A class to define the interaction type for a pair of contact facets.

This is a base class and derived classes define the force generation function when contact between facets occurs.

9.28 point_contact_facet.cc File Reference

Define PointContactFacet functions.

```
#include <cmath>
#include <cstring>
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/numerical.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "../include/contact.hh"
#include "../include/contact_messages.hh"
#include "../include/contact_params.hh"
#include "../include/contact_utils.hh"
#include "../include/point_contact_facet.hh"
#include "../include/point_contact_pair.hh"
```

Namespaces

- jeod

  Namespace jeod.

9.28.1 Detailed Description

Define PointContactFacet functions.

9.29 point_contact_facet.hh File Reference

The contact facet based on the distance to a single point, specifically the vehicle point.

```
#include "dynamics/derived_state/include/relative_derived_state.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "contact_facet.hh"
#include "point_contact_pair.hh"
```

Data Structures

- class jeod::PointContactFacet

  The contact facet based on the distance to a single point, specifically the vehicle point.

Namespaces

- jeod

  Namespace jeod.

9.29.1 Detailed Description

The contact facet based on the distance to a single point, specifically the vehicle point.

In effect this represents a sphere.

9.30 point_contact_facet_factory.cc File Reference

Factory that creates a PointContactFacet from a FlatPlateCircular facet and a ContactParams object.

```
#include <cstddef>
#include <typeinfo>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/surface_model/include/flat_plate_circular.hh"
#include "../include/contact_messages.hh"
#include "../include/contact_params.hh"
#include "../include/point_contact_facet.hh"
#include "../include/point_contact_facet_factory.hh"
```

Namespaces

- jeod

  Namespace jeod.

9.30.1 Detailed Description

Factory that creates a PointContactFacet from a FlatPlateCircular facet and a ContactParams object.

9.31 point_contact_facet_factory.hh File Reference

Creates a point contact facet from an circular flat plate facet.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/surface_model/include/class_declarations.hh"
#include "utils/surface_model/include/interaction_facet.hh"
#include "utils/surface_model/include/interaction_facet_factory.hh"
#include "class_declarations.hh"
#include "point_contact_facet.hh"
```

Data Structures

- class jeod::PointContactFacetFactory

  Creates a PointContactFacet from an InteractionFacet.

Namespaces

- jeod

  Namespace jeod.

9.31.1 Detailed Description

Creates a point contact facet from an circular flat plate facet.

9.32 point_contact_pair.cc File Reference

ContactPair class for use with contact interaction model.

```
#include "utils/named_item/include/named_item.hh"
#include "../include/contact_utils.hh"
#include "../include/pair_interaction.hh"
#include "../include/point_contact_facet.hh"
#include "../include/point_contact_pair.hh"
```

Namespaces

- jeod

  Namespace jeod.

9.32.1 Detailed Description

ContactPair class for use with contact interaction model.

9.33 point_contact_pair.hh File Reference

Class for a pair of point contact facets for use with contact interaction model.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "contact_pair.hh"
```

Data Structures

- class jeod::PointContactPair

    An point to point contact pair for use in the contact model.

Namespaces

- jeod

    Namespace jeod.

9.33.1 Detailed Description

Class for a pair of point contact facets for use with contact interaction model.

9.34 spring_pair_interaction.cc File Reference

spring pair interaction for use in the contact model

```
#include <cmath>
#include "dynamics/derived_state/include/relative_derived_state.hh"
#include "dynamics/dyn_body/include/body_ref_frame.hh"
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "utils/math/include/vector3.hh"
#include "../include/contact_facet.hh"
#include "../include/spring_pair_interaction.hh"
```

Namespaces

- jeod

    Namespace jeod.

9.34.1 Detailed Description

spring pair interaction for use in the contact model

9.35 spring_pair_interaction.hh File Reference

A class for pair interactions based on a simple spring.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "pair_interaction.hh"
```

Data Structures

- class jeod::SpringPairInteraction

  Simple spring contact parameters.

Namespaces

- jeod

  Namespace jeod.

9.35.1 Detailed Description

A class for pair interactions based on a simple spring.

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